

Playing to Strengths: Task Distribution for Heterogeneous Multi-Robot Teams using Monte Carlo Tree Search Guided Dialogue

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Abstract. As robots are deployed in increasingly complex environments, semantic understanding has become essential in enabling them to reason about objects, spatial relationships, and task-relevant context beyond purely geometric representations. Scene graph-based mapping has emerged as a key tool for this, with recent work showing that robots do not need to reason deeply about every part of an environment and focusing detailed inference only on task-relevant regions is sufficient for effective operation. We extend this idea to heterogeneous multi-robot team, where we reframe selective inference as a team-level resource allocation problem rather than a per-robot decision. We propose a framework in which a pair of robots with differing compute capabilities jointly deliberate over semantic observations using a Monte Carlo Tree Search (MCTS) guided actor-critic dialogue, producing delegation plans that respect individual query and travel budgets. We evaluate our approach on a collaborative object retrieval task and show that iterative MCTS dialogue consistently satisfies task constraints and outperforms single-round planning, both in simulation and in real-world experiments.

Keywords: multi-robot systems, semantic reasoning, task delegation

1 Introduction

Robotic systems are rapidly transitioning from single-agent deployments toward heterogeneous multi-robot teams, driven by the need for greater scalability, resilience, and efficiency in complex real-world tasks [1]. As these systems grow in sophistication, semantic understanding has become increasingly important. It enables robots to reason about objects, spatial relationships, and task-relevant context beyond purely geometric representations [2–6]. While semantic reasoning frameworks have matured considerably for single-robot systems, and recent work has begun extending them to multi-robot settings through collaborative scene graph construction [7, 8], multi-robot task planning that leverages semantic observations remains underexplored. In particular, how to coordinate semantic inference across agents with heterogeneous compute capabilities has received little attention.

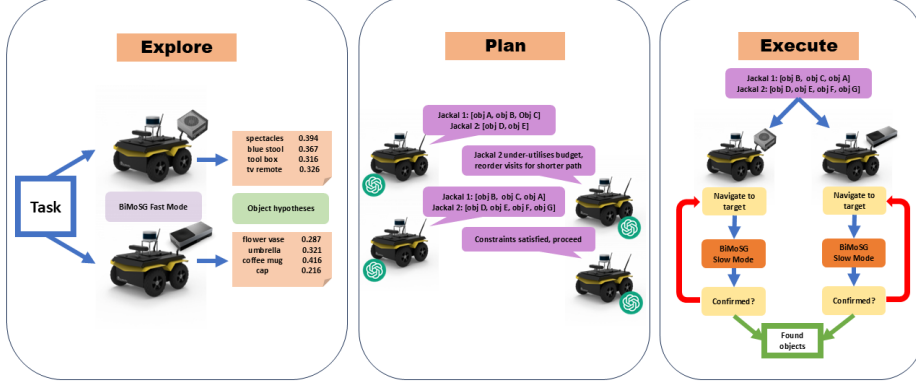


Fig. 1. System overview of our heterogeneous multi-robot framework. The pipeline consists of three phases: semantic exploration using BiMoSG fast mode, MCTS-guided actor-critic task distribution planning, and distributed execution via BiMoSG slow mode verification.

Large foundation models have become the predominant tool for semantic reasoning [9]. Such larger models that allow deep semantic reasoning carry significant computational cost, and robots operating at the edge face strict constraints on onboard resources, latency, and power [10]. This challenge is compounded in heterogeneous multi-robot teams, where individual agents might differ not only in physical capabilities but also in onboard compute capacity. While multi-robot task allocation has been studied extensively [11–13], and semantic inference for single robots has advanced rapidly [3, 4], these two lines of work have largely developed in isolation. The question of how to jointly reason about what to infer and which robot should bear the cost of doing so has not been addressed.

Efforts to reduce the computational burden of semantic inference on robots have largely focused on model-level optimisations [14] rather than on a system level. In our previous work [6], we proposed a bimodal inference framework in which a coarse semantic representation is first constructed, and expensive fine-grained inference is selectively triggered only in task-relevant regions. However, extending this idea to multi-robot systems introduces a fundamentally different problem. What was previously a decision within a single agent becomes a cross-agent coordination problem: given heterogeneous robot capabilities and distributed observations, the team must collectively determine which robot should perform deep inference and on which candidates.

We propose to address this gap by reframing the bimodal inference decision as a team-level resource allocation problem. Rather than asking when a single robot should invoke deep inference, we ask which robot should do so, on which observations, and in what order, given a shared task objective and heterogeneous compute budgets. This reframing exposes a joint optimisation over semantic

confidence, spatial efficiency, and per-robot resource constraints that cannot be resolved by independent per-robot reasoning alone.

To address this, we present a framework in which a team of heterogeneous robots jointly deliberate over semantic observations using a Monte Carlo Tree Search guided actor-critic dialogue. A deterministic scoring function evaluates candidate delegation plans at each step, providing structured feedback that guides the language models toward valid and efficient solutions. An overview of the proposed framework is shown in Fig. 1.

The contributions of this work are as follows:

- A reframing of bimodal semantic inference as a multi-robot resource allocation problem, extending our previous single-agent fast/slow inference paradigm [6] to heterogeneous robot teams.
- An MCTS-guided actor-critic dialogue mechanism for compute-aware task delegation, combining large language models (LLM) reasoning with deterministic plan evaluation.
- Empirical validation on a collaborative object retrieval task, demonstrating that iterative MCTS dialogue consistently produces better delegation plans compared to single-round planning, in both simulation and real-world experiments.

2 Related Work

We situate our work across three related areas: semantic mapping and scene graph generation, multi-robot task allocation and coordination, and structured planning with LLMs.

2.1 Semantic Mapping and Scene Graphs

Scene graphs have emerged as a powerful representation for robotic semantic understanding, encoding objects, their attributes, and spatial relationships in a structured form that supports downstream reasoning and task planning. Early work on 3D scene graphs for robotics [5] demonstrated that hierarchical representations, spanning objects, rooms, and buildings, could be constructed incrementally from sensor streams, enabling robots to reason about their environment at multiple levels of abstraction. Subsequent work has extended this to open-vocabulary settings, where vision-language models allow robots to ground natural language queries against 3D scene representations without requiring a fixed object vocabulary [3, 4]. These approaches have significantly expanded the range of tasks robots can perform through semantic scene understanding [15].

While these scene graph approaches rely on fixed vocabularies or expensive inference for every observation, recent work has explored more efficient strategies. In our previous work [6] we propose a bimodal inference framework for single-robot semantic scene graph generation, combining a lightweight closed-set detector with CLIP [16] for semantic similarity scoring for fast-mode operation,

and selectively triggering a more powerful vision-language model only in regions likely to contain task-relevant objects. This selective usage of larger models reduces unnecessary compute while preserving semantic reasoning quality. However, this decision remains confined to a single agent, and does not account for the coordination challenges that arise when multiple robots with heterogeneous compute capabilities must collectively reason about the same task.

2.2 Multi-Robot Task Allocation and Coordination

Task allocation in multi-robot systems has been studied extensively, with classical approaches relying on auction-based mechanisms [17, 18] and optimisation formulations [19]. These methods have demonstrated effective coordination across a range of multi-robot applications, from search and rescue to warehouse automation.

More recently, LLMs have been applied to multi-robot coordination, leveraging their commonsense reasoning and natural language understanding to produce more flexible task plans. RoCo [20] equips each robot with an LLM agent that engages in dialogue to collectively negotiate task strategies and low-level motion plans, with environment feedback used to iteratively refine proposals in-context. SMART-LLM [21] takes a centralised approach, decomposing high-level instructions into subtasks and allocating them across agents through multi-stage LLM prompting. While these frameworks demonstrate the promise of language-guided coordination for multi-robot systems, RoCo relies on in-context environment feedback rather than an explicit, iterative plan-refinement mechanism, and SMART-LLM relies on a centralised decomposition rather than a decentralised negotiation between agents. Our work addresses both of these gaps with a complete pipeline that first builds an optimised semantic scene graph, uses it to negotiate mission-specific task objectives in a decentralised manner via iterative MCTS-guided dialogue, and finally executes the resulting controls.

2.3 Structured Planning with Large Language Models

The rich semantic knowledge encoded within LLMs also allow them to produce semantically meaningful plans that are grounded in real robot environments. Most notably, SayCan [22] combined the reasoning capabilities of LLMs and learned value functions to select the appropriate skills during task execution. Code as Policies [1] and ProgPrompt [23] uses LLMs to generate high level robot plans in form of executable code. Our work extends the use of LLMs to a multi-robot setup where robots use LLMs to perform task distribution in a decentralized manner. However, we find that LLMs cannot be used out of the box for this purpose, motivating the MCTS-guided dialogue mechanism introduced in this work.

In the context of multi-agent planning, CoELA [24] was proposed as framework for embodied agents to perform cooperative tasks by exchanging information via dialogue. Each agent utilizes LLMs to reason about messages to send to their teammates, and generate execution plans based on the environment as

well as messages from other teammates. CoTS [25] builds upon CoELA by introducing a Monte Carlo Tree Search based collaborative planning module in the planning and communication phase, where each agent acts as either planner or critic to iteratively refine the collaborative plans via MCTS. We adopt the same actor-critic MCTS structure as CoTS, but the two methods differ in how plans are scored. CoTS uses a second LLM call to judge plan quality, while we use a deterministic function based on CLIP confidence, path distance, and budget utilisation, so plan validity can be checked analytically at each node rather than judged by another model. This distinction matters for real-world deployment, where scoring must reflect grounded, physical feasibility rather than an LLM’s subjective judgement of plausibility. The tasks themselves could in principle be generalised with LLMs in either setting. The key distinction we highlight is that we demonstrate our framework in a real-world multi-robot deployment, whereas CoTS has only been demonstrated in simulation.

3 Methodology

3.1 System Overview

Our framework operates in three phases, as illustrated in Fig. 1. During the exploration phase, each robot autonomously navigates its assigned zone and builds a scene graph, together with local object hypotheses using the BiMoSG fast mode pipeline [6], producing CLIP similarity scores and estimated positions for each candidate object. These observations, together with the task objective and various constraints, are passed to the planning module. The planning phase employs an MCTS guided actor-critic dialogue. An actor agent proposes a delegation plan, a deterministic scorer evaluates the plan across three components: (a) CLIP rank quality: whether each robot selects the highest-confidence candidates in its zone; (b) path efficiency: whether the visit ordering minimises total travel distance within each robot’s distance budget; and (c) query budget utilisation: whether each robot fully uses its allocated number of VLM queries. These three components are summed to produce an overall validity score, which is passed to the critic agent alongside the proposed plan. The critic agent provides targeted feedback to guide refinement based on the given information. This loop iterates until a valid plan is found or a preset maximum depth is reached. The resulting plan assigns each robot an ordered list of candidate objects to verify. In the execution phase, each robot navigates to its assigned candidates and triggers the BiMoSG slow mode [6] for verification, contributing confirmed objects toward the team score.

3.2 Bimodal Scene Graph (BiMoSG)

In our previous work [6], we introduced BiMoSG, which generates representations of 3D objects through a relatively low computation cost without the use of large foundation models. This process, dubbed the “fast” mode is able to generate

prismatic object representation of closed set objects through lower cost models. Specifically, given an overall textual goal, BiMoSG uses Mask2Former [26] to generate initial class predictions and CLIP [16] to score how likely each candidate is to fulfil the goal. The robot can then decide to turn on the "slow" mode which utilized VLM to further confirm the target object. This is in contrast to other notable 3D scene graph generation work such as ConceptGraphs [3], which uses vision foundation models directly to generate open-set object representations, typically at higher computational cost, as discussed in our previous work [6].

3.3 Collaborative Task Distribution via Monte Carlo Tree Search

Upon completing the exploration phase, the robots proceed to the task distribution phase, where we introduce a Task Distribution Module that utilises MCTS to iteratively refine delegation plans. Rather than assigning roles to individual robots, two LLM agents, running GPT-4o [27, 28], an actor and a critic jointly deliberate over the shared semantic observations from both zones. The plan proposer takes in the semantic information generated by both robots and proposes an initial plan. The actor proposes an initial delegation plan, which is evaluated by a deterministic scorer and reviewed by the critic. This initial proposal and critique serve as the root node for the planning process, which consists of the following four steps.

(a) Selection: At each iteration, the planning module starts at the root and traverses the tree based on the UCT selection policy until a leaf node is reached. The UCT policy is defined as follows:

$$UCT(c_i) = V(c_i) + w \sqrt{\ln \frac{N(p_i)}{N(c_i)}} \quad (1)$$

where $N(c_i)$ is the number of visits for node c_i , $V(c_i)$ is the back-propagated value of the subtree from c_i , w is the exploration coefficient, and p_i is the parent of c_i .

(b) Expansion: Once a leaf node is reached, the module then expands the tree with n new child nodes. Similar to the root node, each node consists of a distribution plan generated by the proposer and a critique from the critic. Crucially, each child node’s plan is a refinement of the parent node’s plan based on the critique in the parent node.

(c) Evaluation: Each candidate plan is evaluated by a deterministic scorer across three components. (i) Coverage (0–4) rewards query budget utilisation: each robot contributes up to 2 points, scaled by the fraction of its own query budget that is used on valid, in-zone targets. (ii) Rank quality (0–3) rewards selection of high-confidence candidates: for each zone, a softmax-normalised score over the CLIP similarities of the chosen targets contributes up to 1.5 points, summed across the two zones. (iii) Path efficiency (0–3) rewards staying within the distance budget: each robot contributes up to 1.5 points, awarded in full

when its planned path stays within budget and reduced proportionally for each meter travelled beyond it. The three components sum to a raw score out of 10, normalised to $r = \frac{\text{score}}{10} \in [0, 1]$ for backpropagation. A plan is considered valid if it scores $\geq 7.0/10$ and satisfies all hard constraints. The threshold of 7.0 was selected empirically: in pilot trials, plans scoring at or above this threshold consistently satisfied all per-robot hard constraints, while plans scoring below it did not. The hard constraints are: (i) no inference budget violation, meaning a robot does not exceed its allocated number of VLM queries; (ii) no distance budget violation, meaning a robot’s planned path does not exceed its travel budget; (iii) no cross-zone assignment, meaning the planner does not assign a candidate to the robot operating in the opposite zone; and (iv) no duplicate targets, meaning the same candidate does not appear on more than one robot’s list. The critic reads the numerical score and any flagged constraint violations, and provides targeted feedback to the actor to guide the next expansion.

(d) Backpropagation The normalised reward r is back-propagated from the evaluated node to the root. The visitation count and value at each node along the traversal path are updated as follows:

$$N'(c_i) \leftarrow N(c_i) + 1, \quad \forall c_i \in \{c_0, \dots, c_d\} \quad (2)$$

$$V'(c_i) \leftarrow \frac{V(c_i)N(c_i) + r}{N'(c_i)}, \quad \forall c_i \in \{c_0, \dots, c_d\} \quad (3)$$

where d is the current depth of the tree.

Final Plan Selection: The planning module terminates after a preset number of expansions. The final delegation plan is retrieved from the highest-value solved node in the tree. If no solved node exists, the node with the highest back-propagated value $V(c_i)$ is selected as a fallback.

4 Experiments And Results

4.1 Experimental Setup

We designed the experiments around three requirements. First, the two robots needed genuinely different compute capacities, so the delegation problem would actually be non-trivial. Second, the distance budget needed to be tight enough that the planner could not simply pick objects and visit them in any order, and instead had to reason about selection and ordering together. Third, each zone needed a mix of easy and hard detection cases, so the planner’s decisions were not dictated entirely by one extreme or the other.

We conduct experiments in two settings: (a) real-world trials using the physical robot platforms, and (b) simulation trials using a programmatically generated synthetic observations corpus, in which object positions are sampled around randomly placed cluster centers with Gaussian noise, and CLIP similarity scores are drawn from a narrow range calibrated to match the weak rank-to-ground-truth correlation observed in our real-world trials. This allows us to assess performance

at a larger scale than is practical to collect physically, under noise conditions representative of real sensing.

Experiments are conducted in an indoor environment consisting of two circular zones, each approximately 3m in diameter, separated by a physical divider as shown in Fig. 2. Zone 1 is assigned to Jackal 1, a Clearpath Jackal equipped with an NVIDIA Jetson AGX Orin, and Zone 2 to Jackal 2, a Clearpath Jackal equipped with an NVIDIA Jetson AGX Thor. Both robots are equipped with a ZED 2 stereo camera for RGB-D perception.



Fig. 2. Zone 1 is highlighted in green, and Zone 2 is highlighted in orange.

Each zone contains a set of distractor objects arranged around the perimeter, among which four target objects are placed, as shown in Fig. 3. The target objects in Zone 1 are a fire extinguisher, blue stool, toolbox, and television remote. The target objects in Zone 2 are a flower vase, cap, umbrella, and mug. Objects are selected to span a range of detection difficulties. Some are visually distinctive and reliably detected by the fast mode pipeline, while others are smaller or visually similar to distractors as shown in Fig. 4, making them harder to localise with high confidence.

Each robot is allocated an inference budget reflecting its onboard compute capacity. Jackal 1, with lower compute, is permitted at most 2 deep inference

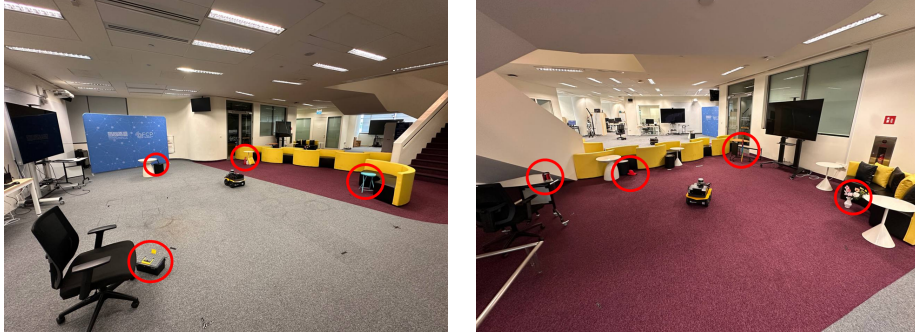


Fig. 3. Experimental zones with target objects circled. Left: Zone 1 (Jackal 1). Right: Zone 2 (Jackal 2).

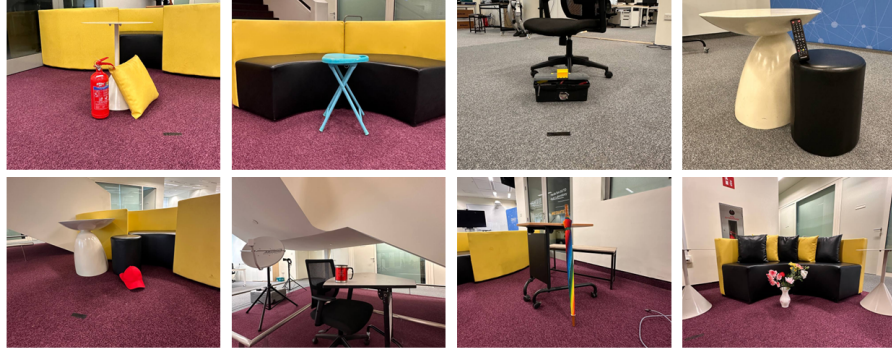


Fig. 4. Target objects used in real-world experiments. Top row: Zone 1 objects (fire extinguisher, blue stool, toolbox, television remote). Bottom row: Zone 2 objects (flower vase, cap, umbrella, mug).

queries per trial, while Jackal 2, with higher compute, is permitted at most 3. This asymmetry is intentional to mirror realistic heterogeneous deployments where more capable agents can shoulder a greater share of expensive inference. A distance budget of 13m per robot per trial is additionally imposed, requiring the planner to consider not only which objects to query but also the order in which they are visited. Together, these constraints encourage the planning module to produce spatially efficient and resource-aware delegation plans rather than naive selections. While the full execution pipeline including physical navigation and VLM-based verification is implemented and operational, we opt to evaluate plan quality against manually verified ground truth rather than physically executing each trial. This allows us to isolate and evaluate the planning mechanism independently of execution noise and physical variability across trials.

4.2 Evaluation Metric

The final delegation plan is evaluated against a ground truth set of correctly detectable objects. For real-world trials, this is established by manually verifying each robot’s scan observations against the physical environment. For simulation trials, ground truth is known by construction from the synthetic data generation process. For each trial, we compute the maximum number of ground truth objects reachable within the inference and distance budgets across all possible visit orderings, which varies per trial depending on the scan observations. A plan is considered successful if the number of confirmed objects equals this maximum, indicating that the delegation plan made optimal use of the available scan observations. Additionally, a plan is considered valid if it meets all of the following conditions: each robot’s query budget is fully utilised, no robot exceeds its distance budget, no object is assigned to the wrong zone, no object appears on both robots’ lists, and the overall plan score meets the validity threshold of $\geq 7.0/10$. We report mean number of objects found, mission success rate, and plan validity across all trials, comparing the initial one-shot plan against the final MCTS-refined plan.

4.3 Baseline

We compare our MCTS-guided dialogue against a single-round planning baseline, in which the actor agent proposes a delegation plan given the same semantic observations, task objective, and constraints, but without any subsequent scoring, critic feedback, or iterative refinement. This corresponds to the initial plan produced at the root node of the MCTS tree before any expansion. All other conditions such as the LLM, prompt structure, and input observations remain identical, ensuring that any difference in plan quality is attributable solely to the iterative MCTS dialogue.

4.4 Experimental Results

We first report results from 10 real-world trials, summarised on a per-trial basis in Table 1. A key finding is that one-shot planning consistently under-utilises the Jackal 2’s higher inference budget in all 10 trials, the initial plan assigned Jackal 2 only 2 objects instead of its full budget of 3, resulting in 0/10 constraint-satisfying plans. This suggests that without iterative feedback, the actor agent defaults to conservative allocations that fail to exploit the heterogeneous compute capabilities of the team. MCTS dialogue reliably corrects this: after refinement, all 10 final plans satisfy the full set of constraints, demonstrating that the critic consistently identifies and addresses budget under-utilisation regardless of trial conditions.

Beyond constraint satisfaction, MCTS-refined plans improved mean number of objects found from 2.4 to 3.1 out of a mean maximum of 3.8. In trials where improvement was observed (trials 1, 3, 6, 8), the revised delegation plan directed Jackal 2 to query a third object that was correctly localised in the scan, directly

Table 1. Per-trial object retrieval results. Objects: number of objects found. Valid: whether the plan satisfied all constraints. Max: maximum reachable objects under distance budget constraints.

Initial (One-Shot)		Final (MCTS)			
Trial	Objects	Valid	Objects	Valid	Max
1	2	×	4	✓	4
2	3	×	3	✓	4
3	2	×	4	✓	5
4	2	×	2	✓	3
5	3	×	3	✓	4
6	2	×	3	✓	3
7	2	×	2	✓	3
8	2	×	4	✓	4
9	4	×	4	✓	4
10	2	×	2	✓	4
Mean	2.4	0/10	3.1	10/10	3.8

contributing to the higher team score. In trials where no improvement was observed despite a valid final plan (trials 2, 4, 5, 7, 10), the limiting factor was scan noise rather than plan quality.

Table 2. MCTS score convergence across 10 real-world trials.

Planning step	Mean score (/10)	Constraint-satisfied
Initial (one-shot)	8.98	0/10
Expand 1	9.63	7/10
Expand 2	9.75	7/10
Expand 3	9.75	7/10
Expand 4	9.81	8/10
Expand 5+	9.81	8/10

Table 2 shows the convergence of MCTS planning scores across 10 trials. The most significant improvement occurs at the first expansion, where constraint-satisfied plans jump from 0/10 to 7/10 and mean score increases from 8.98 to 9.63. Beyond expand 1, gains diminish rapidly. Constraint satisfaction reaches 8/10 at expand 4 and plateaus thereafter, with mean score stabilising at 9.81. This suggests that for problems of this scale, 4 to 5 MCTS expansions are sufficient to produce near-optimal delegation plans, and further expansions yield negligible improvement. The remaining 2/10 trials that never achieve constraint satisfaction correspond to cases where scan noise produced candidate positions that made it geometrically infeasible to visit 3 objects within the distance budget, representing a hard limit imposed by the environment rather than the planner.

Table 3. Summary of simulation experiment results.

Metric	Initial (one-shot)	Final (MCTS)
Mean no. of objects found	4.80	6.40
Mean max possible		7.75
Mean improvement		+1.60
Retrieval efficiency	61.9%	82.6%
Valid plan	0/20	17/20

Table 3 presents the results of the simulation experiments, conducted on a scaled-up setting with 10 objects per zone and asymmetric query budgets of 4 and 8 for Jackal 1 and Jackal 2 respectively, reflecting a more pronounced degree of compute heterogeneity. Despite the increased combinatorial complexity, MCTS-refined plans consistently outperform one-shot planning, improving mean number of objects found from 4.80 to 6.40 and retrieval efficiency from 61.9% to 82.6%. Constraint satisfaction improves from 0/20 to 17/20, consistent with the real-world finding that one-shot planning reliably under-utilises the higher-compute agent’s query budget. The remaining 3 trials that failed to satisfy constraints correspond to cases where the spatial distribution of candidate objects made it geometrically difficult to visit the full query budget within the distance budget, representing a hard limit imposed by the environment rather than the planner.

4.5 Discussion

The results across both real-world and simulation experiments point to a consistent finding: one-shot LLM planning reliably under-utilises the higher-compute agent’s query budget, producing constraint-violating plans in all trials regardless of problem scale. This suggests that without iterative feedback, LLMs default to conservative allocations that fail to exploit heterogeneous compute capabilities. This limitation is not obvious from the “raw objects-found” metric alone, but becomes clear when constraint satisfaction is tracked explicitly. MCTS dialogue addresses this directly, with the critic consistently identifying and correcting budget under-utilisation in the first expansion. The similarity in retrieval efficiency improvement across real-world ($\sim 18\%$) and simulation ($\sim 21\%$) settings further suggests that this correction mechanism generalises beyond the specific object set and environment used in physical experiments.

In the real-world experiments, the gap between final and maximum possible objects found is partly attributable to upstream perception quality. The exploration phase initialises the planner using CLIP similarity scores computed over closed-set detection. This is an approach we are able to rely on directly because our framework, unlike prior multi-robot coordination work, spans the full pipeline from perception through to planning and control. When detections are incorrect, for instance a fire extinguisher detected as a bottle, or a stool detected

as a chair, the planner operates on imperfect observations and has no mechanism to recover ground truth objects that were never correctly identified during exploration. In trials where no improvement was observed despite a valid final plan, the remaining gap is largely attributable to the tightness of the combined query and distance constraints. With a fixed budget, the planner cannot visit all high-confidence candidates, and in some spatial configurations no delegation strategy can recover the remaining ground truth objects. These cases represent the fundamental limits of budget-constrained planning rather than a shortcoming of the MCTS dialogue itself. The planner produces the best feasible plan given the constraints, but the constraints themselves prevent a perfect outcome.

This work demonstrates that iterative multi-agent dialogue is beneficial for task distribution in heterogeneous robot teams. The consistent improvement in constraint satisfaction and objects found across both real-world and simulation experiments validates the core premise that structured LLM dialogue, grounded by a deterministic scorer, can produce better delegation plans than single-round planning. Future work will explore scaling this decentralised framework to teams of more than two robots, to fully demonstrate its decentralisation capabilities. The underlying combination of iterative dialogue and deterministic evaluation also extends naturally to other multi-robot tasks that require reasoning about a shared, finite resource, such as collaborative exploration, dynamic re-planning, and scaling to larger heterogeneous teams with uncertainty-aware scoring.

5 Conclusion

We presented a framework for heterogeneous multi-robot task delegation that reframes selective bimodal inference as a team-level resource allocation problem. By introducing an MCTS-guided actor-critic dialogue grounded by a deterministic scorer, we enable a pair of robots with differing compute capabilities to jointly deliberate over semantic observations and produce constraint-satisfying delegation plans. Experiments across both real-world and simulation settings demonstrate that iterative MCTS dialogue consistently outperforms single-round planning in terms of constraint satisfaction and objects retrieved, validating the premise that structured multi-agent dialogue is beneficial for task distribution in heterogeneous robot teams. Across both settings, MCTS dialogue improved retrieval efficiency by approximately 18–21% over single-round planning, suggesting that structured collective reasoning over inference resources is a practical design principle for heterogeneous robot teams.

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